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Open-Minded

Higgs bundles on stacky curves

Anna Borri
Doctoral Researcher (10/2024-now)

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Motivation

For a smooth projective complex curve X, the non-abelian Hodge correspondence can be understood as relating the geometry of the generic fiber and the special fiber inside the moduli space of λ-connections on X

$$\begin{tikzcd}
\mathcal{M}^0_{n,Dol}(X) \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{M}^0_{n,Hdg}(X) \arrow[d] & \mathcal{M}^0_{n,DR}(X) \arrow[l, hook'] \arrow[r, "\cong"] \arrow[d] \arrow[dl, phantom, "\llcorner"] & \mathcal{M}^0_{n,B}(X) = Loc_n(X) \\
\{0\} \arrow[r] & \mathbb{A}^1 & \{1\} \arrow[l] &
\end{tikzcd}$$

Problem. Degree d vector bundles don't admit flat connection. How to get a good notion of $\mathcal{M}^d_{n,DR}(X)$ that fits into a similar description?

The stacky curve $\mathcal{X}$

Fix a point $p \in X$. We define the stacky curve $\mathcal{X}$ to be the n-th root stack of X at p

$$\begin{tikzcd}
\mathcal{X} = \sqrt[n]{p/X} \arrow[r] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner"] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge n"] \\
X \arrow[r, "(\mathcal{O}(p), s_p)"'] & {[\mathbb{A}^1/\mathbb{G}_m]}.
\end{tikzcd}$$

Degree d on a curve as degree 0 on $\mathcal{X}$

Vector bundles on $\mathcal{X}$ have a third numerical invariant beyond rank and degree, namely the representation obtained as stalk at the stacky point. We denote by $\lambda(-d)$ the weight $-d$ character of μ_n and by $n \cdot \lambda(-d)$ the corresponding n-dimensional representation.

Theorem. The twisted pullback and pushforward along the moduli map $\mathcal{X} \to X$ define equivalences of algebraic stacks

$$\mathcal{M}^d_{n,Dol}(X) \cong \mathcal{M}^0_{n\cdot\lambda(-d),Dol}(\mathcal{X}).$$

Upshot. Deegree d Higgs bundles on X can be understood as degree 0 Higgs bundles on $\mathcal{X}$.

Twisted character variety as Betti moduli space on $\mathcal{X}$

Computing the fundamental group of $\mathcal{X}$, we see that the Betti moduli space on $\mathcal{X}$ is

$$\mathcal{M}_{n,B}(\mathcal{X}) = \left[\left\{((A_i, B_i)_{i=1}^g, C) \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = C,\ C^n = Id\right\}/GL_n\right].$$

It decomposes with respect to the conjugacy class of C, and the component corresponding to $\mathcal{M}^0_{n\cdot\lambda(-d),Dol}(\mathcal{X})$ is

$$\mathcal{M}_{n\cdot\lambda(-d),B}(\mathcal{X}) = \left[\left\{(A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = e^{-\frac{2\pi i}{n}d} Id\right\}/GL_n\right].$$

This is exactly the twisted character variety of X.

Upshot. The twisted character variety of X is a connected component of the character variety of $\mathcal{X}$.

Consequence

We can identify $\mathcal{M}^d_{n,DR}(X)$ with $\mathcal{M}^0_{n\cdot\lambda(-d),DR}(\mathcal{X})$. Therefore, the inclusions $\mathcal{M}_{n,Dol}(\mathcal{X}) \hookrightarrow \mathcal{M}_{n,Hdg}(\mathcal{X})$ and $\mathcal{M}_{n,DR}(\mathcal{X}) \hookrightarrow \mathcal{M}_{n,Hdg}(\mathcal{X})$ and the (analytic) Riemann-Hilbert correspondence $\mathcal{M}_{n,DR}(\mathcal{X}) \cong \mathcal{M}_{n,B}(\mathcal{X})$ induce isomorphisms in cohomology

$$H^*(\mathcal{M}^d_{n,Dol}(X)) \cong H^*(\mathcal{M}^d_{n,DR}(X)) \cong H^*(\mathcal{M}^d_{n,B}(X)).$$

Upshot. The non-abelian Hodge correspondence for coprime n, d on X can be understood from the non-abelian Hodge correspondence for 0-degree bundles on $\mathcal{X}$.

What about the Hodge structure?

The equivalence between the De Rham and the Betti moduli space is analytic, so it doesn't preserve the Hodge structure.

Theorem. The tautological class $e_{i,j}$ (i.e. the j-th Künneth component of $c_i(\mathcal{E}_{univ})$) on $\mathcal{M}_{n,Dol}(\mathcal{X})$ has pure weight j. Under the isomorphism $\mathcal{M}_{n,DR}(\mathcal{X}) \cong \mathcal{M}_{n,B}(\mathcal{X})$ it is mapped to a cohomology class of weight $2i$.

Computing the Hodge weights

To compute the Hodge weights we replicate Shende's argument presented in [1]. We construct a simplicial set $\Delta_{\mathcal{X}}$ realizing a triangulation of $\mathcal{X}$. We make it into a simplicial scheme and use the equivalence

$$\mathcal{M}_{n,B}(\mathcal{X}) = [\mathrm{Hom}(\pi_1(\mathcal{X}, v_0), GL_n)/GL_n] \cong \mathcal{H}om_{SSch}(\Delta_X, BGL_n)$$

and the universal object on $\mathcal{H}om_{SSch}(\Delta_X, BGL_n)$ to carry out the computation of the weights.

References

[1] Vivek Shende. "The weights of the tautological classes of character varieties". In: *Int. Math. Res. Not. IMRN* 22 (2017), pp. 6832–6840. ISSN: 1073-7928,1687-0247. DOI: 10.1093/imrn/rnv363. URL: https://doi.org/10.1093/imrn/rnv363.

[2] Carlos Simpson. "Local systems on proper algebraic V-manifolds". In: *Pure Appl. Math. Q.* 7.4 (2011), pp. 1675–1759. ISSN: 1558-8599,1558-8602. DOI: 10.4310/PAMQ.2011.v7.n4.a27. URL: https://doi.org/10.4310/PAMQ.2011.v7.n4.a27.

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